

Milestone Gears Pvt. Ltd., 4, Opp. Himachal Fibers Ltd. Area, Barotiwala, Distt. - (HP)				Standard Operating Procedure				OPN. NO.	पिछला सं. / रंश	Shot Blasting	MACHINE NO.	Lab.																																																	
								220	अगला आपरेशन	Gear Chamfering	REVISION NO.	00																																																	
								REVISION DATE																																																					
								DATE OF ENACTMENT																																																					
								25/11/13																																																					
4th GEAR - C/MESH 1512A01501				CUST. NAME				APPD. BY: <i>[Signature]</i>																																																					
MS/SOP/3055/220				MATERIAL				TOOLS / CUTTERS DETAIL :																																																					
SECTION INSTRUMENTS DETAIL :				TAFE(8x8)				OPERATION																																																					
				16MnCr5				CHD. BY: <i>[Signature]</i>																																																					
				VALUE				FIXTURE NUMBER :																																																					
				MACHINE / EQUIPMENT PARAMETERS				CLAMP ☐																																																					
				VALUE				REST ☐																																																					
				JIG / FIX. / CTG. TOOL PARAMETERS				INSPECT ☒																																																					
				VALUE				MAJOR MJ																																																					
								MINOR Mn																																																					
								CRITICAL Cr																																																					
								DO & DON'T S.																																																					
								1 सभी Instrument साफ सुथरे होने चाहिये।																																																					
								2 सभी Instrument का Leveling सही होना चाहिये।																																																					
								3 सभी Instrument Calibrate होने चाहिये।																																																					
								CONTROLLED COPY SR. NO. 01 ISSUED TO <i>[Signature]</i> DATED 22/11/21																																																					
								NOTE:- मशीन चलाने से पहले आप्रैटर पिछला आपरेशन चैक करेगा। पिछला आपरेशन होने के बाद ही अगला आपरेशन करेगा।																																																					
				OPTN. SKETCH																																																									
				CORE HARDNESS MEASURED AT ROOT DIA 6.4 FROM END OF TOOTH ECD & HARDNESS MEASURED AT MIDPOINT OF TOOTH PROFILE 6.4 FROM END OF TOOTH																																																									
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				Mettallurgical Testing के लिये नीचे लिखे Instruction को Follow करें :- 1) Rockwell Hardness Tester में Component की Surface Hardness चैक करें। 2) Component को काट कर एक सैम्पल चैक करें। 3) Rockwell/Vickers Hardness Tester में Sample की Core Hardness चैक करें। 4) Vickers Hardness Tester में Sample की ECD. चैक करें। 5) Nital से Sample को Etching करें। 6) Etching किये हुये Sample का Case & Core में Microstructure Microscope द्वारा चैक करें।																																																									
				CONTROLLING OF DIEMENSION																																																									
				<table border="1"> <thead> <tr> <th>No</th> <th>Dimension</th> <th>Specifications</th> <th>Instrument</th> <th>Operator</th> <th>Inspector</th> <th>Freq.</th> </tr> </thead> <tbody> <tr> <td>A.</td> <td>Previous Operation Surface Hardness</td> <td>Shot Blasting</td> <td>Visually</td> <td>100%</td> <td>100%</td> <td>1 piece per heat</td> </tr> <tr> <td>1.</td> <td>ECD (Cut off HRC)</td> <td>58~63HRC</td> <td>Rockwell Tester</td> <td>--do--</td> <td>--do--</td> <td>--do--</td> </tr> <tr> <td>2.</td> <td>Core Hardness</td> <td>0.76~1.14mm</td> <td>Vicker H. Tester</td> <td>--do--</td> <td>--do--</td> <td>--do--</td> </tr> <tr> <td>3.</td> <td>Surface Carbon Content</td> <td>27~41HRC</td> <td>Rockwell Tester</td> <td>--do--</td> <td>--do--</td> <td>--do--</td> </tr> <tr> <td>4.</td> <td>Surface Hardness on Internal Bore</td> <td>0.80~1.10 %</td> <td>Microscope</td> <td>--do--</td> <td>--do--</td> <td>--do--</td> </tr> <tr> <td>5.</td> <td></td> <td>27HRC Min.</td> <td>Rockwell Tester</td> <td>--do--</td> <td>--do--</td> <td>--do--</td> </tr> </tbody> </table>				No	Dimension	Specifications	Instrument	Operator	Inspector	Freq.	A.	Previous Operation Surface Hardness	Shot Blasting	Visually	100%	100%	1 piece per heat	1.	ECD (Cut off HRC)	58~63HRC	Rockwell Tester	--do--	--do--	--do--	2.	Core Hardness	0.76~1.14mm	Vicker H. Tester	--do--	--do--	--do--	3.	Surface Carbon Content	27~41HRC	Rockwell Tester	--do--	--do--	--do--	4.	Surface Hardness on Internal Bore	0.80~1.10 %	Microscope	--do--	--do--	--do--	5.		27HRC Min.	Rockwell Tester	--do--	--do--	--do--					
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				Rejection होने पर Machine को तुरन्त रोक दें और shift incharge और Q.C. Insp. को मदद से SPOT ANALYSIS कराएँ। उसके बाद ही Machine को चलाएँ।																																																									
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